

Edge Length Loss for Improved Building Polygon Extraction

KUANREN QIAN, Amazon.com, USA

MOHAMED MOUSTAFA, Amazon.com, USA

The automated generation of accurate polygonal building footprints from overhead imagery is crucial for numerous geospatial applications. While recent deep learning models excel at extracting polygon from images, the predicted outlines often exhibit an overabundance of short, unnecessary edge segments, creating visually unrealistic and hyper-crenelated outlines that diminish practical applications despite potential metric advantages. To solve this issue, we introduce a novel, fully differentiable edge length loss (ELL) designed to regularize excessively short edges based on “blurry” vertex coordinates derived from the output logits. This “blurry” coordinate approach offers robustness against common ground truth misalignment, promoting geometric plausibility while enabling end-to-end training with geometric regularization. Through integration into the Pix2Poly framework, we demonstrate that this loss encourages the generation of smoother, more geometrically plausible building outlines while improving the performance of the core sequence prediction mechanism across almost all metrics.

CCS Concepts: • **Computing methodologies** → **Object detection; Computer vision representations; Neural networks; Machine learning**; • **Information systems** → **Geographic information systems**.

Additional Key Words and Phrases: Building Footprint Extraction, Geometric Regularization, Deep Learning, Remote Sensing, Edge length loss, Hyper-crenelation

ACM Reference Format:

Kuanren Qian and Mohamed Moustafa. 2025. Edge Length Loss for Improved Building Polygon Extraction. In *Proceedings of the 4th ACM SIGSPATIAL International Workshop on Spatial Big Data and AI for Industrial Applications (GeoIndustry '25)*, November 3–6, 2025, Minneapolis, MN, USA. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/nnnnnnn.nnnnnnn>

1 INTRODUCTION

Building footprints extraction from remote sensing data, such as aerial and satellite imagery, is a foundational task in applications from geospatial analysis and urban planning to city modeling and change detection [9, 12]. The preferred output format is typically precise vector polygons that faithfully delineate building perimeters. However, conventional methodologies relying on pixel-wise semantic segmentation often necessitate intricate post-processing routines to transform raster outputs into vector polygons. These derived polygons frequently suffer from geometric imperfections, such as rounded corners or irregular edge structures, and may not capture the inherent regularity of man-made structures [10].

In tackle these limitations, end-to-end deep learning architectures that directly regress polygonal representations have gained significant traction. Among them, Pix2Poly [2] has emerged as a notable advancement, conceptualizing polygon extraction as a sequence prediction problem using a transformer-based model generates an list of vertex coordinates. Although Pix2Poly demonstrates proficiency in producing high-fidelity building footprints, its predictions, similar to other direct

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

GeoIndustry '25, November 3–6, 2025, Minneapolis, MN, USA

© 2025 Association for Computing Machinery.

ACM ISBN 978-1-4503-XXXX-X/YY/MM...\$15.00

<https://doi.org/10.1145/nnnnnnn.nnnnnnn>

methods, can sometimes be marred by an undesirable prevalence of jagged edge segments, leading to “hyper-crenelated” boundaries, which inefficiently represent typical building morphologies and complicates downstream applications.

To directly address the geometric deficiencies in predicted building polygons, such as hyper-crenelation and misaligned ground truth vertices, we introduce a novel differentiable edge length loss (ELL) based on “blurry” coordinates. This regularization term is engineered to enhance geometric quality by systematically penalizing the formation of excessively short edges. The central innovation of ELL lies in its computation based on “blurry” vertex coordinates, a probabilistic interpretation of the output logits. This formulation ensures the entire regularization mechanism is differentiable, seamlessly integrating into end-to-end training paradigms.

The main contributions of this work are:

- A novel, fully differentiable edge length loss (ELL) that regularizes building polygon geometry by penalizing spurious short edges while maintaining end-to-end trainability;
- A “blurry” coordinate formulation that encodes model confidence, enabling image-aware geometric regularization, contrary to geometry-only post-processing; and
- Comprehensive evaluation demonstrating that ELL not only produces smoother, more plausible building outlines but also improves core detection metrics across multiple datasets.

2 RELATED WORK

2.1 Raster Segmentation to Vector Polygons

The automated extraction of building footprints has witnessed a significant evolution, transitioning from classical image processing to sophisticated deep learning models. Majority work address building footprint extraction as a semantic segmentation problem [5, 6]. These methods employ fully convolutional networks or transformer-based architectures to generate pixel-wise building masks. While achieving high intersection over union (IoU) on segmentations, the raster outputs requires additional vectorization procedures, often involving contour detection followed by polygonal approximations. Such two-stage processes can introduce geometric inaccuracies, failing to preserve sharp corners or rectilinear edges inherent in building structures. Efforts to improve vector quality from masks include specialized loss functions or regularization applied during segmentation or post-processing [8, 22].

2.2 End-to-End Polygon Prediction

Early efforts to circumvent raster-to-vector conversion for polygon generation utilized Recurrent neural networks for autoregressive vertex sequence prediction, such as Polygon-RNN++ [1]. More recently, transformer-based architectures such as PolyFormer [14], and P2PFormer [23] have shown superior performance in directly generating polygon sequences or converting learned primitives. Graph neural networks have also been applied to learn vertex representations and connectivity [24]. A key advancement in direct prediction method is the Pix2Poly framework [2], which formulates building footprint extraction as an image-to-sequence task using a vision transformer encoder and a transformer decoder, followed by an optimal matching network for connectivity, enabling end-to-end differentiability. Other notable direct approaches include BuildMapper [19], the work by [4], and techniques that integrate semantic and geometric learning pathways [13] with recent foundation models like SAM [11] and its derivatives SAMPolyBuild [18].

2.3 Geometric Regularity and Plausibility

Ensuring geometric integrity and plausibility in direct polygon generation is a persistent challenge. Models like Pix2Poly [2], while aiming for high fidelity, often yield “jagged” or hyper-crenelated

outlines from an overabundance of short edges, where, analogous to the coastline paradox, increased vertex density can diminish rather than enhance real-world utilization. Such vertex-dense polygons may achieve high scores on fine-grained metrics, yet prove impractical or even misleading in geospatial and business applications by introducing unnecessary complications for downstream use, underscoring the critical need for robust regularization evident in work on incorporating existing geometric information [8] and outline refinement [22]. Our proposed ELL contributes a targeted, differentiable strategy that specifically discourages undesirable short segments by operating on “blurry” vertex coordinates, thereby promoting smoother, more practically realistic building footprints within architectures like Pix2Poly.

3 METHODOLOGY

We designed a differentiable edge length loss to enhance building outline prediction and outline plausibility, based on the Pix2Poly model. This section first provides a brief overview of the Pix2Poly architecture and then presents a detailed formulation for ELL.

3.1 Pix2Poly Revisited

The Pix2Poly architecture [2] extracts building footprints end-to-end using two main components and a combined loss function:

- **Vertex Sequence Detector:** This module encodes the input image with a vision transformer to generate latent features. Ground truth vertex coordinates (x, y) are quantized into discrete integer tokens, forming sequences with special tokens for start, end, and separation of polygons. A transformer decoder then autoregressively predicts vertex coordinate tokens, based on the image features and preceding tokens.
- **Optimal Matching Network:** Using features from the detector, this model predicts a permutation matrix for clockwise and counter-clockwise connectivity, and forms closed polygons using a differentiable matching algorithm.
- **Combined loss function:**

$$\mathcal{L}_{\text{Pix2Poly}} = \lambda_s \cdot \mathcal{L}_{\text{detection}}^v + \lambda_p \cdot \mathcal{L}_{\text{permutation}} \quad (1)$$

where $\mathcal{L}_{\text{detection}}^v$ and $\mathcal{L}_{\text{permutation}}$ are the cross-entropy losses for the predicted vertex token sequence and the connectivity matrix, respectively, balanced by weights λ_s and λ_p [2].

3.2 Differentiable Edge Length Loss (ELL)

To enhance geometric practicality and mitigate “jagged” outlines from excessive short edges in predicted building polygons, we introduce a differentiable edge length loss (ELL) that penalize short segments and operates on “blurry” vertex coordinates derived from the output logits (Algorithm 1).

The “blurry” coordinates are computed as follows: (1) Apply softmax to logits ($\mathbf{P}_{\text{logits}}$) over N_{bins} spatial bins for each coordinate; (2) Calculate “blurry” coordinates ($\mathbf{C}_{\text{norm}}$) using a weighted sum of bin centers and their probabilities; and (3) Scale $\mathbf{C}_{\text{norm}}$ by image dimensions to obtain pixel-space positions ($\mathbf{C}_{\text{scaled}}$). This “blurry” coordinate represents the expected normalized position, derived from predicted probability distributions across discrete spatial bins. It naturally encodes model confidence, where confident predictions yield precise positions, while uncertainty leads to intermediary “blurry” coordinates, implicitly handling ambiguity. This differentiable process enables end-to-end training and provides robustness against ground truth misalignment.

Using $\mathbf{C}_{\text{scaled}}$, ELL computes edge lengths (L_i) of cyclically connected edges using Euclidean distance. It applies a squared penalty, $\max(0, L_{\text{edge}} - L_i)^2$, to edges shorter than L_{edge} , strongly discouraging unnecessary short edge segments. $\mathcal{L}_{\text{ELL}}$ is the mean of penalties. Its differentiability

Algorithm 1 Differentiable Edge Length Loss

Input: logits P , image dimensions (H, W) , edge threshold L_{edge} , vertex count N_v , bin count N_b
Output: edge length loss $\mathcal{L}_{ELL}$

```

/* Calculate Expected Vertex Coordinates */
1:  $P_{coord} = P[:, : 2N_v, : N_b]$                                 ▶ Extract coordinate bins
2:  $P_{prob} = \text{Softmax}(P_{coord})$                                 ▶ Apply softmax
3:  $v_{centers} = [k / (N_b - 1) \text{ for } k \in 0..N_b - 1]$         ▶ Bin centers
4:  $P_{reshaped} = \text{Reshape}(P_{prob}, (\text{batch}, N_v, 2, N_b))$ 
5:  $C_{norm} = \sum_k (P_{reshaped}[:, :, :, k] \cdot v_{centers}[k])$     ▶ Normalization
6:  $C_{scaled} = C_{norm} \cdot [H, W]$                                 ▶ Scale to image dimensions
/* Compute edge length penalties */
7: for  $i = 0$  to  $N_v - 1$  do
8:    $p_1 = C_{scaled}[:, i, :]$                                 ▶ Current vertex
9:    $p_2 = C_{scaled}[:, (i + 1) \bmod N_v, :]$                 ▶ Next vertex
10:   $L = \sqrt{\sum (p_2 - p_1)^2 + \epsilon}$                             ▶ Edge length
11:  penalty append  $\max(0, L_{edge} - L_i)^2$ 
12: end for
13:  $\mathcal{L}_{ELL} = \text{Mean}(\text{penalties})$ 
14: return  $\mathcal{L}_{ELL}$ 

```

allows direct regularization of the vertex sequence detector during end-to-end training, guiding the model towards smoother, geometrically practical building outlines.

Our approach (ELL) can be integrated into the Pix2Poly framework via combined loss function:

$$\mathcal{L}_{\text{total}} = (\lambda_s \cdot \mathcal{L}_{\text{detection}}^v + \lambda_p \cdot \mathcal{L}_{\text{permutation}}) + \lambda_{ELL} \cdot \mathcal{L}_{ELL} \quad (2)$$

where λ_{ELL} weights the geometric regularization. This combined loss guides the model to predict accurate vertex sequences and connectivity while ensuring plausible geometric properties, avoiding an overabundance of short, noisy edges.

4 EXPERIMENTAL RESULTS

4.1 Dataset

We evaluated ELL on two datasets: INRIA aerial images [16] and CrowdAI [17]. Our analysis shows INRIA contains simpler buildings with relatively regular outlines, while CrowdAI features more complex ground truth polygons. This difference makes them ideal for testing ELL’s effectiveness on different boundary complexity. For the INRIA dataset, we follow pre-processing steps as specified in [2], obtaining 112,896 image-mask pair for training and 16,896 for validation. For the experiments presented in this paper, we utilized two randomly sampled CrowdAI subsets: a 3% subset consists of 8,422 training images and 1,810 validation images, and a 10% subset consists of 28,074 training images and 6,032 validation images.

4.2 Implementation Details

ViT-Small [7] with a patch size of 8 is used as the backbone. Training was conducted for 500 epochs with a batch size of 12, using the AdamW optimizer [15] with a learning rate of 4×10^{-4} and weight decay of 1×10^{-4} . Following [2], the loss weights were set to $\lambda_s = 1.0$ for the detection loss, and $\lambda_p = 10.0$ for the permutation loss. The ELL weight, λ_{ELL} , was empirically set to 0.01 based on loss values, with a minimum edge length threshold (L_{edge}) of 5.0 pixels. Coordinate quantization used $N_{bins} = 224$ bins, and ($N_{vertices}$) as 256. All models were trained on 8 NVIDIA V100-16 GPUs.

4.3 Quantitative Comparison

Table 1. Polygonal footprints quality metrics comparison.

Method	Dataset	IoU↑	$N_{ratio} = 1$	PoLiS↓	IoU _{topo} ↑	F _{topo} ↑	PA _{topo} ↑
Pix2Poly ELL(ours)	3% CrowdAI	55.1 60.2	1.0181 1.0174	2.1945 2.1493	43.86 47.32	59.04 62.54	92.25 92.64
Pix2Poly ELL(ours)	10% CrowdAI	69.2 77.2	1.0834 0.9671	2.0084 1.8934	49.65 55.34	64.26 70.10	93.13 94.10
Pix2Poly ELL(ours)	INRIA	79.38 79.96	1.0851 1.0754	1.9317 1.936	61.05 60.58	74.32 73.86	94.35 95.52

Note: Metrics differ from [2] due to hyperparameter variations not detailed in public repositories. We refer readers to [2] for complete STOA metrics.

Table 1 shows ELL improves building polygon geometry and detection accuracy on INRIA and CrowdAI datasets, including COCO metrics, IoU, PoLiS [3], and topological metrics [21]. ELL shows improvements across all metrics, but here we only include selected metrics due to space constraints. On 10% CrowdAI dataset, ELL enhances the PoLiS score by 5.7% to 1.8934 and achieves a more ideal N_{ratio} of 0.9671, while improving mAP by 1.6 to 32.2 and IoU by 8.0 to 77.2. Similar improvement, including a 4.9% mAP increase on the 3% subset, showcasing ELL’s effectiveness. By penalizing short edges based on “blurry” coordinates, this approach yields smoother polygons that capture the actual building geometry and better overall detection. Table 2 benchmarks our results against SOTA shown in [2]. Our Pix2Poly baseline deviated from the original, likely due to difference in undisclosed experimental setups, such as large tile splitting and patching. Despite that, ELL improved the original Pix2Poly’s performance [2].

Table 2. Quantitative comparison with SOTA models.

Method	Weight	IoU	Accuracy
Frame Field Learning (FFL) [8]	76.6M	74.80%	95.96%
HiSup [20]	74.3M	75.53%	96.27%
Pix2Poly [2]	31.9M	75.87%	96.37%
Pix2Poly(reproduce)	31.9M	72.38%	95.91%
ELL(ours)	31.9M	73.05%	96.02%

Note: Evaluation on INRIA official test split.

4.4 Qualitative Improvements

Figure 1 demonstrates the effect of ELL, corresponding to the quantitative metrics. Comparing to the often hyper-crenelated results from the baseline Pix2Poly, our edge length loss produces smoother, more plausible building outlines with less crenelation, demonstrating effective mitigation of noisy short segments. Visualization reveals that edge length loss consistently produces polygons with fewer micro-segments and more plausible right angles compared to the baseline. Additionally, by penalizing closely-spaced vertices, ELL naturally prevents overlapping duplicate predictions (Figure 1a top), further contributing to cleaner building outlines.

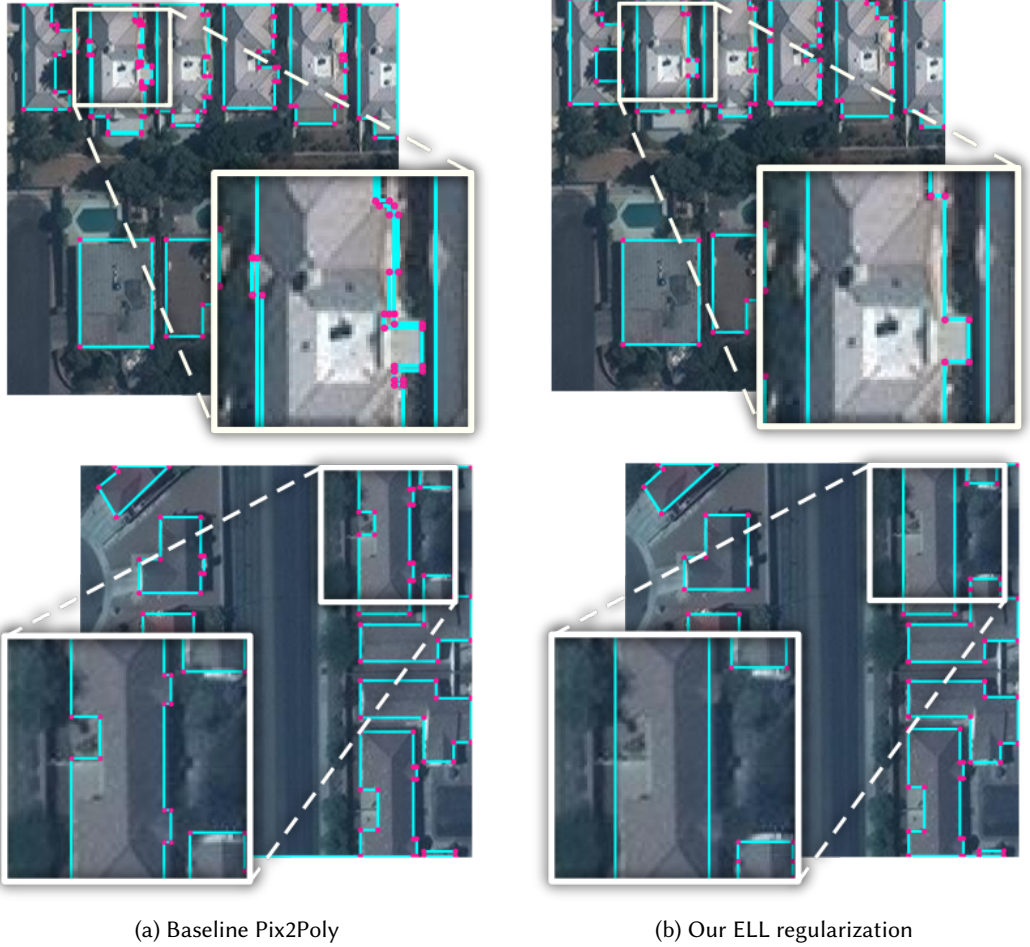


Fig. 1. Comparison using two CrowdAI samples. (Magenta dots) Detected vertices. (Cyan lines) Building outlines.

4.5 Discussion of Results

Tables 1 and Figure 1 demonstrate that the our ELL approach effectively mitigates hyper-crenelation and enhances the geometric plausibility of extracted building polygons. This is evidenced by consistent improvements in geometric metrics such as PoLiS and N_{ratio} across both the 3% and 10% CrowdAI dataset subsets, alongside visually smoother outputs. The mechanism behind this improvement lies in ELL, which penalizes short edges calculated from "blurry" vertex coordinates. This encourages the training to focus on more distant consecutive vertices, leading to more regularized outlines.

Interestingly, our approach not only refines outlines but also yields notable gains in core detection and segmentation accuracy, as seen in the improved IoU, topological scores (Table 1). While not detailed here, positive trends were observed across COCO metrics. The "blurry" coordinate encourages the model to explore vertex predictions in challenging scenarios and handle partially

obscured buildings, promoting the model to capture more geometrically sound representations and offering a differentiable target for outline regularization, particularly for ground truth misalignment.

The differential impact of ELL, more pronounced on CrowdAI than INRIA, stems directly from distinct ground truth characteristics. CrowdAI features an overabundance of short edges (99.68% of buildings, 1.86 average short edges/building, only 0.3% with none), leading to significant gains with ELL. INRIA has smoother annotations (90.99% affected, 1.50 average short edges/building, 9.0% with none), limiting correction scope by ELL and thus explaining its more moderate, yet reasonable, effect. This highlights ELL’s strength in regularizing noisy short edges prevalent in data.

5 CONCLUSION AND FUTURE WORK

This paper introduced a novel differentiable edge length loss (ELL) designed to enhance the geometric quality of building polygons generated by sequence-based prediction models, specifically targeting the issue of hyper-crenelation. By operating on “blurry” vertex coordinates derived from model logits, our approach provides a differentiable regularization that effectively discourages excessively short edges, leading to smoother and more practical building outlines. Future work will prioritize evaluations of ELL across diverse datasets to confirm generalization. We will also explore adaptive edge length thresholds, geometric constraints, LLM-based footprint generation, and improving imperfect ground truth.

ACKNOWLEDGMENTS

REFERENCES

- [1] David Acuna, Huan Ling, Amlan Kar, and Sanja Fidler. 2018. Efficient interactive annotation of segmentation datasets with polygon-RNN++. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 859–868.
- [2] Yeshwanth Kumar Adimoolam, Charalambos Poullis, and Melinos Averkiou. 2025. Pix2Poly: A Sequence Prediction Method for End-to-end Polygonal Building Footprint Extraction from Remote Sensing Imagery. In *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. IEEE, 8484–8493.
- [3] Janja Avbelj, Rupert Müller, and Richard Bamler. 2014. A metric for polygon comparison and building extraction evaluation. *IEEE Geoscience and Remote Sensing Letters* 12, 1 (2014), 170–174.
- [4] Qi Chen, Lei Wang, Steven L Waslander, and Xiuguo Liu. 2020. An end-to-end shape modeling framework for vectorized building outline generation from aerial images. *ISPRS Journal of Photogrammetry and Remote Sensing* 170 (2020), 114–126.
- [5] Bowen Cheng, Ishan Misra, Alexander G Schwing, Alexander Kirillov, and Rohit Girdhar. 2022. Masked-attention mask transformer for universal image segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 1290–1299.
- [6] Bowen Cheng, Alex Schwing, and Alexander Kirillov. 2021. Per-pixel classification is not all you need for semantic segmentation. *Advances in Neural Information Processing Systems* 34 (2021), 17864–17875.
- [7] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. 2020. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929* (2020).
- [8] Nicolas Girard, Dmitriy Smirnov, Justin Solomon, and Yuliya Tarabalka. 2021. Polygonal building extraction by frame field learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 5891–5900.
- [9] Yang He, Emre Eftelioglu, Mohamed Moustafa, and Amber Roy Chowdhury. 2023. A Highly Efficient and Effective Attribute Learning Framework for Road Graph from Aerial Imagery and GPS. In *Proceedings of the 11th ACM SIGSPATIAL International Workshop on Analytics for Big Geospatial Data*. 32–41.
- [10] Weiqin Jiao, Claudio Persello, and George Vosselman. 2024. PolyR-CNN: R-CNN for end-to-end polygonal building outline extraction. *ISPRS Journal of Photogrammetry and Remote Sensing* 218 (2024), 33–43.
- [11] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C Berg, Wan-Yen Lo, et al. 2023. Segment anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 4015–4026.
- [12] Qingyu Li, Lichao Mou, Yao Sun, Yuansheng Hua, Yilei Shi, and Xiao Xiang Zhu. 2024. A review of building extraction from remote sensing imagery: Geometrical structures and semantic attributes. *IEEE Transactions on Geoscience and Remote Sensing* (2024).

- [13] Weijia Li, Wenqian Zhao, Huaping Zhong, Conghui He, and Dahua Lin. 2021. Joint semantic-geometric learning for polygonal building segmentation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 35. 1958–1965.
- [14] Jiang Liu, Hui Ding, Zhaowei Cai, Yuting Zhang, Ravi Kumar Satzoda, Vijay Mahadevan, and R Manmatha. 2023. PolyFormer: Referring image segmentation as sequential polygon generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 18653–18663.
- [15] Ilya Loshchilov and Frank Hutter. 2017. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101* (2017).
- [16] Emmanuel Maggiori, Yuliya Tarabalka, Guillaume Charpiat, and Pierre Alliez. 2017. Can semantic labeling methods generalize to any city? the INRIA aerial image labeling benchmark. In *2017 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*. IEEE, 3226–3229.
- [17] Sharada Prasanna Mohanty, Jakub Czakon, Kamil A Kaczmarek, Andrzej Pyskir, Piotr Tarasiewicz, Saket Kunwar, Janick Rohrbach, Dave Luo, Manjunath Prasad, Sascha Fleer, et al. 2020. Deep Learning for Understanding Satellite Imagery: An Experimental Survey. *Frontiers in Artificial Intelligence* 3 (2020).
- [18] Chenhao Wang, Jingbo Chen, Yu Meng, Yupeng Deng, Kai Li, and Yunlong Kong. 2024. SAMPolyBuild: Adapting the segment anything model for polygonal building extraction. *ISPRS Journal of Photogrammetry and Remote Sensing* 218 (2024), 707–720.
- [19] Shiqing Wei, Tao Zhang, Shunping Ji, Muying Luo, and Jianya Gong. 2023. BuildMapper: A fully learnable framework for vectorized building contour extraction. *ISPRS Journal of Photogrammetry and Remote Sensing* 197 (2023), 87–104.
- [20] Bowen Xu, Jiakun Xu, Nan Xue, and Gui-Song Xia. 2023. HiSup: Accurate polygonal mapping of buildings in satellite imagery with hierarchical supervision. *ISPRS Journal of Photogrammetry and Remote Sensing* 198 (2023), 284–296.
- [21] Bingnan Yang, Mi Zhang, Zhan Zhang, Zhili Zhang, and Xiangyun Hu. 2023. TopDiG: Class-agnostic topological directional graph extraction from remote sensing images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 1265–1274.
- [22] Min Yang, Renwei Zou, Tinghua Ai, and Xiongfeng Yan. 2024. Regularizing building outlines extracted from remote sensing images by integrating multiple algorithms. *Geocarto International* 39, 1 (2024), 2370322.
- [23] Tao Zhang, Shiqing Wei, Yikang Zhou, Muying Luo, Wenling Yu, and Shunping Ji. 2024. P2PFormer: A Primitive-to-polygon Method for Regular Building Contour Extraction from Remote Sensing Images. *IEEE Transactions on Geoscience and Remote Sensing* (2024).
- [24] Stefano Zorzi, Shabab Bazrafkan, Stefan Habenschuss, and Friedrich Fraundorfer. 2022. PolyWorld: Polygonal building extraction with graph neural networks in satellite images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 1848–1857.